

PROJECT DESIGN PHASE

Problem – Solution Fit

Date	30 September 2026
Team ID	SWTID-2026-3024
Project Name	AI-Augmented Backend Development AI BlogNest API
Maximum Marks	2 Marks

1. CUSTOMER SEGMENT(S)

- Bloggers and content creators
 - Authors and writers
 - Editors and content managers
 - Users who regularly create digital content
 - Organizations managing multiple blog posts
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2. JOBS-TO-BE-DONE / PROBLEMS

- Create blog content efficiently.
 - Manage multiple blog posts in one place.
 - Generate content ideas and drafts quickly.
 - Create summaries for lengthy content.
 - Manage users, comments, and blog information.
 - Maintain secure access to content.
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3. TRIGGERS

- Need to create a new blog article.
 - Large amount of content needs to be managed.
 - Manual writing takes too much time.
 - Need to summarize lengthy articles.
 - Need for secure and centralized content management.
 - Need for AI assistance during content creation.
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4. EMOTIONS: BEFORE / AFTER

Before:

- Time-consuming
- Frustrated with repetitive writing
- Difficulty managing multiple posts
- Concern about organizing content

After:

- Faster content creation
 - Reduced manual effort
 - Better content organization
 - More convenient content management
 - Greater confidence in secure access
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5. CUSTOMER CONSTRAINTS

- Limited time for creating content.
 - Manual content creation requires considerable effort.
 - Managing multiple blog posts can become difficult.
 - Users may have different access requirements.
 - Security of user accounts and content must be maintained.
 - AI-generated content may require review and editing.
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6. AVAILABLE SOLUTIONS

- Traditional blogging platforms
 - Manual content creation tools
 - Content management systems
 - Online AI writing tools
 - Separate summarization tools
 - Manual methods for managing blog content
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7. BEHAVIOUR

- Creates and updates blog posts.
 - Searches for topics and content ideas.
 - Reviews and edits content.
 - Organizes blog information.
 - Manages comments and user activities.
 - Uses digital tools to improve content creation.
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8. CHANNELS OF BEHAVIOUR

8.1 ONLINE

- Web applications
- REST APIs
- AI-powered content services
- Online content management platforms
- Postman for API interaction and testing

8.2 OFFLINE

- Planning blog topics
- Writing and reviewing content
- Discussing content requirements with team members
- Reviewing and editing generated content

9. PROBLEM ROOT CAUSE

The major root cause is the dependence on manual processes for creating, organizing, and managing blog content. Traditional systems may not provide integrated AI assistance, centralized content management, secure role-based access, and automated content summarization in a single backend platform.

10. YOUR SOLUTION

AI BlogNest is an AI-powered blog content management system that combines RESTful APIs, MongoDB, secure authentication, role-based access, blog management, and Artificial Intelligence.

The system helps users generate blog content, summarize existing content, manage blog posts and comments, and securely manage user access through a centralized backend platform.